CS 229, Fall 2018
Problem Set #1: Supervised Learning

1. [40 points] Linear Classifiers (logistic regression and GDA)
In this problem, we cover two probabilistic linear classifiers we have covered in class so far. First, a discriminative linear classifier: logistic regression. Second, a generative linear classifier: Gaussian discriminant analysis (GDA). Both algorithms find a linear decision boundary that separates the data into two classes, but make different assumptions. Our goal in this problem is to get a deeper understanding of the similarities and differences (and strengths and weaknesses) of these two algorithms.
For this problem, we will consider two datasets, provided in the following files:
   i. data/ds1_{train,valid}.csv
   ii. data/ds2_{train,valid}.csv

   Each file contains m examples, one example $(x^{(i)}, y^{(i)})$ per row. In particular, the ith row contains columns $x_0^{(i)} \in \mathbb{R}$, $x_1^{(i)} \in \mathbb{R}$, and $y^{(i)} \in \{0, 1\}$. In the subproblems that follow, we will investigate using logistic regression and Gaussian discriminant analysis (GDA) to perform binary classification on these two datasets.

   (a) [10 points] In lecture we saw the average empirical loss for logistic regression:

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(h_\theta(x^{(i)})\right) + (1-y^{(i)})\log\left(1-h_\theta(x^{(i)})\right)\right],$$

   where

$$y^{(i)} \in \{0,1\}, \qquad h_\theta(x) = g(\theta^T x), \qquad g(z) = \frac{1}{1+e^{-z}}.$$

   Find the Hessian H of this function, and show that for any vector z, it holds true that

$$z^T H z \geq 0.$$

   Hint: You may want to start by showing that

$$\sum_i \sum_j z_i x_i x_j z_j = (x^T z)^2 \geq 0.$$

   Recall also that

$$g'(z) = g(z)(1-g(z)).$$

   Remark: This is one of the standard ways of showing that the matrix H is positive semi-definite, written $H \succeq 0$. This implies that J is convex, and has no local minima other than the global one. If you have some other way of showing $H \succeq 0$, you are also welcome to use your method instead of the one above.

   Solution.
   Let

$$h_i := h_\theta(x^{(i)}) = g(\theta^T x^{(i)}).$$

   For the ith training example, define

$$\ell_i(\theta) = -\left[y^{(i)}\log h_i + (1-y^{(i)})\log(1-h_i)\right].$$

   Then

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m}\ell_i(\theta).$$

   Since

$$g'(t) = g(t)(1-g(t)),$$

   we have

$$\frac{\partial h_i}{\partial \theta_j} = h_i(1-h_i)x_j^{(i)}.$$

   Therefore,

$$\begin{aligned}\frac{\partial \ell_i}{\partial \theta_j} &= -\frac{y^{(i)}}{h_i}\frac{\partial h_i}{\partial \theta_j} + \frac{1-y^{(i)}}{1-h_i}\frac{\partial h_i}{\partial \theta_j} \\ &= -y^{(i)}(1-h_i)x_j^{(i)} + (1-y^{(i)})h_i x_j^{(i)} \\ &= (h_i - y^{(i)})x_j^{(i)}.\end{aligned}$$

Hence,

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{m}\sum_{i=1}^{m}(h_i - y^{(i)})x_j^{(i)},$$

and thus

$$\nabla J(\theta) = \frac{1}{m}\sum_{i=1}^{m}(h_i - y^{(i)})x^{(i)}.$$

Now consider the (j,k) entry of the Hessian:

$$\begin{aligned} H_{jk} &= \frac{\partial^2 J}{\partial \theta_j \partial \theta_k} \\ &= \frac{1}{m}\sum_{i=1}^{m}\frac{\partial h_i}{\partial \theta_k}x_j^{(i)} \\ &= \frac{1}{m}\sum_{i=1}^{m}h_i(1-h_i)x_j^{(i)}x_k^{(i)}. \end{aligned}$$

Therefore,

$$\boxed{H = \frac{1}{m}\sum_{i=1}^{m}h_i(1-h_i)x^{(i)}(x^{(i)})^T}.$$

To show that H is positive semidefinite, let $z \in \mathbb{R}^n$ be arbitrary. Then

$$\begin{aligned} z^T H z &= \frac{1}{m}\sum_{i=1}^{m}h_i(1-h_i)z^T x^{(i)}(x^{(i)})^T z \\ &= \frac{1}{m}\sum_{i=1}^{m}h_i(1-h_i)\left((x^{(i)})^T z\right)^2. \end{aligned}$$

Since

$$0 < h_i < 1,$$

we have

$$h_i(1-h_i) \geq 0,$$

and clearly

$$\left((x^{(i)})^T z\right)^2 \geq 0.$$

Thus every term in the summation is nonnegative, so

$$\boxed{z^T H z \geq 0}$$

for every $z \in \mathbb{R}^n$.
Hence,

$$\boxed{H \succeq 0}.$$

(b) [5 points] Coding problem.
Follow the instructions in src/p01b_logreg.py to train a logistic regression classifier using Newton's Method. Starting with

$$\theta = \vec{0},$$

run Newton's Method until the updates to θ are small. Specifically, train until the first iteration k such that

$$\|\theta_k - \theta_{k-1}\|_1 < \epsilon, \qquad \epsilon = 1 \times 10^{-5}.$$

Make sure to write your model's predictions to the file specified in the code.

(c) [5 points]
Recall that in GDA we model the joint distribution of (x,y) by the following equations:

$$p(y) = \begin{cases} \phi, & y = 1, \\ 1-\phi, & y = 0, \end{cases}$$

$$p(x \mid y=0) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_0)^T\Sigma^{-1}(x-\mu_0)\right),$$

and

$$p(x \mid y=1) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)\right),$$

where ϕ, μ_0, μ_1, and Σ are the parameters of our model.
Suppose we have already fit ϕ, μ_0, μ_1, and Σ, and now want to predict y given a new point x.
To show that GDA results in a classifier that has a linear decision boundary, show the posterior distribution can be written as

$$p(y=1 \mid x; \phi, \mu_0, \mu_1, \Sigma) = \frac{1}{1+\exp\left(-(\theta^T x + \theta_0)\right)},$$

where

$$\theta \in \mathbb{R}^n, \qquad \theta_0 \in \mathbb{R}$$

are appropriate functions of ϕ, Σ, μ_0, and μ_1.

Solution.
By Bayes' rule,

$$\begin{aligned} p(y=1 \mid x) &= \frac{p(x \mid y=1)P(y=1)}{p(x)} \\ &= \frac{p(x \mid y=1)P(y=1)}{p(x \mid y=1)P(y=1) + p(x \mid y=0)P(y=0)}. \end{aligned}$$

Substituting the Gaussian densities, we obtain

$$\begin{aligned} p(y=1 \mid x) = \frac{\phi \exp\left(-\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)\right)}{\phi \exp\left(-\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)\right)} \\ + (1-\phi)\exp\left(-\frac{1}{2}(x-\mu_0)^T\Sigma^{-1}(x-\mu_0)\right). \end{aligned}$$

Dividing the numerator and denominator by the numerator,

$$p(y=1 \mid x) = \frac{1}{1+\frac{1-\phi}{\phi}\exp\left(-\frac{1}{2}\left[(x-\mu_0)^T\Sigma^{-1}(x-\mu_0) - (x-\mu_1)^T\Sigma^{-1}(x-\mu_1)\right]\right)}.$$

Let $\Omega = \Sigma^{-1}$. Since Σ is symmetric, its inverse Ω is also symmetric.
Now consider the difference of the quadratic forms:

$$\begin{aligned} &(x-\mu_0)^T\Omega(x-\mu_0) - (x-\mu_1)^T\Omega(x-\mu_1) \\ &= (x^T-\mu_0^T)\Omega(x-\mu_0) - (x^T-\mu_1^T)\Omega(x-\mu_1) \\ &= x^T\Omega x - x^T\Omega\mu_0 - \mu_0^T\Omega x + \mu_0^T\Omega\mu_0 \\ &\quad - x^T\Omega x + x^T\Omega\mu_1 + \mu_1^T\Omega x - \mu_1^T\Omega\mu_1. \end{aligned}$$

Using the symmetry of Ω, we have

$$\mu_i^T\Omega x = x^T\Omega\mu_i.$$

Hence,

$$\begin{aligned} &(x-\mu_0)^T\Omega(x-\mu_0) - (x-\mu_1)^T\Omega(x-\mu_1) \\ &= 2x^T\Omega(\mu_1-\mu_0) + \mu_0^T\Omega\mu_0 - \mu_1^T\Omega\mu_1. \end{aligned}$$

Substituting this expression back into the posterior,

$$p(y=1 \mid x) = \frac{1}{1+\frac{1-\phi}{\phi}\exp\left(-x^T\Omega(\mu_1-\mu_0) - \frac{1}{2}\mu_0^T\Omega\mu_0 + \frac{1}{2}\mu_1^T\Omega\mu_1\right)}.$$

Since

$$\frac{1-\phi}{\phi} = \exp\left(-\log\frac{\phi}{1-\phi}\right),$$

we may rewrite the posterior as

$$p(y=1 \mid x) = \frac{1}{1+\exp\left(-\left[x^T\Omega(\mu_1-\mu_0) + \frac{1}{2}\mu_0^T\Omega\mu_0 - \frac{1}{2}\mu_1^T\Omega\mu_1 + \log\frac{\phi}{1-\phi}\right]\right)}.$$

Therefore, defining

$$\boxed{\theta = \Sigma^{-1}(\mu_1-\mu_0)}$$

and by using the differnece-of-squares identity:

$$\boxed{\theta_0 = \frac{1}{2}\mu_0^T\Sigma^{-1}\mu_0 - \frac{1}{2}\mu_1^T\Sigma^{-1}\mu_1 + \log\frac{\phi}{1-\phi}, = \log\frac{\phi}{1-\phi} - \frac{1}{2}(\mu_0+\mu_1)^T\theta}$$

we conclude that

$$\boxed{p(y=1 \mid x) = \frac{1}{1+\exp(-(\theta^T x+\theta_0))}.}$$

□

(d) [7 points]
In this problem, we derive the maximum likelihood estimators for Gaussian Discriminant Analysis (GDA), first in one dimension and then in arbitrary finite dimensions.
Let

$$\mathcal{D} = \left\{(x^{(i)}, y^{(i)})\right\}_{i=1}^m$$

be a dataset consisting of m independent observations, where $x^{(i)} \in \mathbb{R}^n$ and $y^{(i)} \in \{0,1\}$.
Recall that under the GDA model,

$$p(y^{(i)};\phi) = \phi^{y^{(i)}}(1-\phi)^{1-y^{(i)}},$$

and

$$X \mid Y=k \sim \mathcal{N}(\mu_k, \Sigma), \qquad k \in \{0,1\}.$$

The log-likelihood of the observations is

$$\ell(\phi,\mu_0,\mu_1,\Sigma) = \log\prod_{i=1}^m p(x^{(i)} \mid y^{(i)};\mu_0,\mu_1,\Sigma)p(y^{(i)};\phi).$$

You may assume that the dataset contains at least one positive and one negative example.

(i) The One-Dimensional Case.
Suppose first that $n=1$, so that

$$\mu_0,\mu_1 \in \mathbb{R}, \qquad \Sigma = [\sigma^2], \qquad \sigma^2 > 0.$$

By expanding the log-likelihood and applying ordinary differentiation, show that the maximum likelihood estimators of ϕ and μ_k are given by

$$\boxed{\hat{\phi} = \frac{1}{m}\sum_{i=1}^m \mathbf{1}y^{(i)} = 1}$$

and

$$\boxed{\hat{\mu}_k = \frac{\sum_{i=1}^m \mathbf{1}y^{(i)} = kx^{(i)}}{\sum_{i=1}^m \mathbf{1}y^{(i)} = k}, \qquad k \in \{0,1\}.}$$

Now write $v = \sigma^2$ and define

$$s = \frac{1}{m}\sum_{i=1}^m \left(x^{(i)} - \hat{\mu}_{y^{(i)}}\right)^2.$$

Show that, after maximizing over the mean parameters, the terms in the log-likelihood which depend on v can be expressed as

$$\ell(v) = -\frac{m}{2}\left(\log v + \frac{s}{v}\right) + \text{constant}.$$

Assuming $s > 0$, determine the value of v which maximizes this expression, and hence prove that

$$\boxed{\hat{\sigma}^2 = \frac{1}{m}\sum_{i=1}^{m}\left(x^{(i)} - \hat{\mu}_{y^{(i)}}\right)^2.}$$

(ii) Generalization to Arbitrary Dimensions.
Suppose now that n is an arbitrary positive integer, so that

$$x^{(i)}, \mu_0, \mu_1 \in \mathbb{R}^n, \qquad \Sigma \in \mathbb{R}^{n\times n},$$

where Σ is symmetric positive definite.
By choosing an appropriate change of variable, define the precision matrix

$$\boxed{K = ?.}$$

Show that the log-likelihood can be expressed as

$$\begin{aligned}\ell(\phi, \mu_0, \mu_1, K) = &\sum_{i=1}^{m}\left[y^{(i)}\log\phi + (1-y^{(i)})\log(1-\phi)\right] \\ &+ \frac{m}{2}\log\det K \\ &- \frac{1}{2}\sum_{i=1}^{m}(x^{(i)} - \mu_{y^{(i)}})^T K (x^{(i)} - \mu_{y^{(i)}}) \\ &+ \text{constant}.\end{aligned}$$

For $k \in \{0, 1\}$, define

$$m_k = \sum_{i=1}^{m} \mathbf{1}y^{(i)} = k.$$

By differentiating the log-likelihood with respect to the mean vectors, show that

$$\boxed{\nabla_{\mu_k}\ell = K\left(\sum_{i:y^{(i)}=k} x^{(i)} - m_k\mu_k\right).}$$

Hence, by applying the first-order optimality condition, deduce that

$$\boxed{\hat{\mu}_k = \frac{\sum_{i=1}^{m}\mathbf{1}y^{(i)} = kx^{(i)}}{\sum_{i=1}^{m}\mathbf{1}y^{(i)} = k}, \qquad k \in \{0, 1\}.}$$

Verify that these stationary points maximize the log-likelihood with respect to the mean parameters.

Now define the residual vectors

$$r_i = x^{(i)} - \hat{\mu}_{y^{(i)}}$$

and the scatter matrix

$$\boxed{S = \sum_{i=1}^{m} r_i r_i^T.}$$

Assume that S is positive definite.
By expressing the quadratic terms using the trace of a matrix, show that the log-likelihood, as a function of K, can be written as

$$\boxed{\ell(K) = \frac{m}{2}\log\det K - \frac{1}{2}\operatorname{tr}(KS) + \text{constant}.}$$

Recall the following matrix differentiation identities:

$$\nabla_A \operatorname{tr}(AB) = B^T,$$

and, for an invertible matrix A,

$$\nabla_A \det A = \det(A)(A^{-1})^T.$$

By applying the chain rule, deduce that for a symmetric positive definite matrix A,

$$\boxed{\nabla_A \log\det A = (A^{-1})^T.}$$

Hence, by differentiating the log-likelihood with respect to the precision matrix K, show that

$$\boxed{\nabla_K \ell = \frac{1}{2}\left(mK^{-1} - S\right).}$$

By applying the first-order optimality condition, show that

$$K^{-1} = \frac{1}{m}S.$$

You may use the fact that $\log\det K$ is strictly concave on the set of symmetric positive definite matrices. Explain why this implies that the stationary point is the unique global maximizer of the log-likelihood. Finally, recalling that $K = \Sigma^{-1}$, deduce that

$$\boxed{\hat{\Sigma} = \frac{1}{m}\sum_{i=1}^{m}\left(x^{(i)} - \hat{\mu}_{y^{(i)}}\right)\left(x^{(i)} - \hat{\mu}_{y^{(i)}}\right)^T.}$$

Conclude that the maximum likelihood estimators obtained in part (i) extend to arbitrary finite dimensions, provided that the empirical residual covariance matrix is positive definite.

Solution.

(i) The One-Dimensional Case.

Suppose that $n = 1$, so that $\mu_0, \mu_1 \in \mathbb{R}$ and $\Sigma = [\sigma^2]$.
Write $v = \sigma^2 > 0$.
By the assumptions of GDA,

$$p(y^{(i)}; \phi) = \phi^{y^{(i)}}(1-\phi)^{1-y^{(i)}},$$

and

$$p(x^{(i)} \mid y^{(i)}) = \frac{1}{\sqrt{2\pi v}}\exp\left(-\frac{(x^{(i)} - \mu_{y^{(i)}})^2}{2v}\right).$$

Since the training observations are independent, the log-likelihood is

$$\begin{aligned}
\ell(\phi, \mu_0, \mu_1, v) &= \sum_{i=1}^{m}\log\left[p(x^{(i)} \mid y^{(i)})p(y^{(i)})\right] \\
&= \sum_{i=1}^{m}\left[y^{(i)}\log\phi + (1-y^{(i)})\log(1-\phi)\right] \\
&\quad - \frac{m}{2}\log(2\pi) - \frac{m}{2}\log v \\
&\quad - \frac{1}{2v}\sum_{i=1}^{m}(x^{(i)} - \mu_{y^{(i)}})^2.
\end{aligned}$$

We maximize the log-likelihood with respect to ϕ, μ_0, μ_1, and v.

Estimation of ϕ.

Define

$$m_k = \sum_{i=1}^{m}\mathbf{1}y^{(i)} = k, \qquad k \in \{0, 1\},$$

so that $m_0 + m_1 = m$.
The terms depending on ϕ are

$$\ell_\phi = m_1\log\phi + m_0\log(1-\phi).$$

Differentiating with respect to ϕ,

$$\frac{\partial \ell}{\partial \phi} = \frac{m_1}{\phi} - \frac{m_0}{1-\phi}.$$

By the first-order optimality condition,

$$\begin{aligned} \frac{m_1}{\phi} - \frac{m_0}{1-\phi} &= 0, \\ m_1(1-\phi) &= m_0\phi, \\ m_1 &= (m_0 + m_1)\phi = m\phi. \end{aligned}$$

Therefore,

$$\boxed{\hat{\phi} = \frac{m_1}{m} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}y^{(i)} = 1.}$$

Moreover,

$$\frac{\partial^2 \ell}{\partial \phi^2} = -\frac{m_1}{\phi^2} - \frac{m_0}{(1-\phi)^2} < 0.$$

Hence, the log-likelihood is strictly concave in ϕ, and the stationary point is its unique global maximum.

Estimation of μ_0 and μ_1.

For $k \in \{0, 1\}$, the terms depending on μ_k are

$$\ell_{\mu_k} = -\frac{1}{2v} \sum_{i:y^{(i)}=k} (x^{(i)} - \mu_k)^2.$$

Differentiating with respect to μ_k,

$$\begin{aligned} \frac{\partial \ell}{\partial \mu_k} &= -\frac{1}{2v} \sum_{i:y^{(i)}=k} 2(x^{(i)} - \mu_k)(-1) \\ &= \frac{1}{v} \sum_{i:y^{(i)}=k} (x^{(i)} - \mu_k). \end{aligned}$$

Setting the derivative equal to zero,

$$\begin{aligned} \sum_{i:y^{(i)}=k} (x^{(i)} - \mu_k) &= 0, \\ \sum_{i:y^{(i)}=k} x^{(i)} &= m_k\mu_k. \end{aligned}$$

Thus,

$$\boxed{\hat{\mu}_k = \frac{1}{m_k} \sum_{i:y^{(i)}=k} x^{(i)} = \frac{\sum_{i=1}^{m} \mathbf{1}y^{(i)} = kx^{(i)}}{\sum_{i=1}^{m} \mathbf{1}y^{(i)} = k}.}$$

Furthermore,

$$\frac{\partial^2 \ell}{\partial \mu_k^2} = -\frac{m_k}{v} < 0.$$

Therefore, each stationary point is the unique maximizer with respect to μ_k.
Notice that the optimal mean parameters do not depend on the value of v.

Estimation of σ^2.

Substituting the maximum likelihood estimates of the mean parameters, define

$$s = \frac{1}{m}\sum_{i=1}^{m}(x^{(i)} - \hat{\mu}_{y^{(i)}})^2.$$

The terms depending on v become

$$\begin{aligned} \ell(v) &= -\frac{m}{2}\log v - \frac{1}{2v}\sum_{i=1}^{m}(x^{(i)} - \hat{\mu}_{y^{(i)}})^2 + C \\ &= -\frac{m}{2}\left(\log v + \frac{s}{v}\right) + C, \end{aligned}$$

where C is independent of v.
Differentiating,

$$\ell'(v) = -\frac{m}{2}\left(\frac{1}{v} - \frac{s}{v^2}\right)$$
$$= \frac{m(s-v)}{2v^2}.$$

The first-order optimality condition gives

$$\ell'(v) = 0 \quad \Longleftrightarrow \quad v = s.$$

Since $s > 0$ and $v > 0$,

$$\ell'(v) \begin{cases} > 0, & 0 < v < s, \\ = 0, & v = s, \\ < 0, & v > s. \end{cases}$$

Hence, $v = s$ is the unique global maximizer.
Recalling that $v = \sigma^2$, we obtain

$$\boxed{\hat{\sigma}^2 = \frac{1}{m}\sum_{i=1}^{m}(x^{(i)} - \hat{\mu}_{y^{(i)}})^2.}$$

This completes part (i).

(ii) Generalization to Arbitrary Dimensions.
Now suppose that n is an arbitrary positive integer, with

$$x^{(i)}, \mu_0, \mu_1 \in \mathbb{R}^n, \qquad \Sigma \in \mathbb{R}^{n\times n}, \qquad \Sigma \succ 0.$$

Make a change of variable: define the precision matrix

$$\boxed{K = \Sigma^{-1}.}$$

Since Σ is symmetric positive definite, K is also symmetric positive definite. Note that K and Σ^{-1} is bijective, hence optimizing l w.r.t. Σ^{-1} is equivalent of optimizing l w.r.t. K

The Multivariate Log-Likelihood.
Recall that

$$p(x^{(i)} \mid y^{(i)}) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x^{(i)} - \mu_{y^{(i)}})^T\Sigma^{-1}(x^{(i)} - \mu_{y^{(i)}})\right).$$

Taking logarithms and substituting $K = \Sigma^{-1}$, we obtain

$$\ell(\phi, \mu_0, \mu_1, K) = \sum_{i=1}^{m}\Big[y^{(i)}\log\phi + (1-y^{(i)})\log(1-\phi)\Big]$$
$$- \frac{mn}{2}\log(2\pi) - \frac{m}{2}\log\det\Sigma$$
$$- \frac{1}{2}\sum_{i=1}^{m}(x^{(i)} - \mu_{y^{(i)}})^T K(x^{(i)} - \mu_{y^{(i)}}).$$

Since

$$\log\det\Sigma = -\log\det K,$$

we may rewrite this as

$$\boxed{\begin{aligned}\ell(\phi, \mu_0, \mu_1, K) = &\sum_{i=1}^{m}\Big[y^{(i)}\log\phi + (1-y^{(i)})\log(1-\phi)\Big] \\ &+ \frac{m}{2}\log\det K \\ &- \frac{1}{2}\sum_{i=1}^{m}(x^{(i)} - \mu_{y^{(i)}})^T K(x^{(i)} - \mu_{y^{(i)}}) + C,\end{aligned}}$$

where $C = -\frac{mn}{2}\log(2\pi)$.
The terms involving ϕ are identical to those in part (i). Therefore,

$$\boxed{\hat{\phi} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}y^{(i)} = 1.}$$

Estimation of the Mean Vectors.
For $k \in \{0, 1\}$, consider the terms depending on μ_k:

$$\ell_{\mu_k} = -\frac{1}{2}\sum_{i:y^{(i)}=k} (x^{(i)} - \mu_k)^T K (x^{(i)} - \mu_k).$$

Recall that for a symmetric matrix A,

$$\nabla_u \left[(x-u)^T A (x-u)\right] = 2A(u-x).$$

Since K is symmetric,

$$\begin{aligned}
\nabla_{\mu_k}\ell &= -\frac{1}{2}\sum_{i:y^{(i)}=k} 2K(\mu_k - x^{(i)}) \\
&= K\sum_{i:y^{(i)}=k} (x^{(i)} - \mu_k) \\
&= K\left(\sum_{i:y^{(i)}=k} x^{(i)} - m_k\mu_k\right).
\end{aligned}$$

Hence,

$$\boxed{\nabla_{\mu_k}\ell = K\left(\sum_{i:y^{(i)}=k} x^{(i)} - m_k\mu_k\right).}$$

Applying the first-order optimality condition,

$$K\left(\sum_{i:y^{(i)}=k} x^{(i)} - m_k\mu_k\right) = 0.$$

Since $K \succ 0$, it is invertible. Therefore,

$$\sum_{i:y^{(i)}=k} x^{(i)} - m_k\mu_k = 0.$$

Consequently,

$$\boxed{\hat{\mu}_k = \frac{\sum_{i=1}^{m} \mathbf{1}y^{(i)} = kx^{(i)}}{\sum_{i=1}^{m} \mathbf{1}y^{(i)} = k}, \qquad k \in \{0, 1\}.}$$

Moreover, the Hessian with respect to μ_k is

$$\boxed{\nabla^2_{\mu_k}\ell = -m_k K.}$$

Since $m_k > 0$ and $K \succ 0$,

$$-m_k K \prec 0.$$

Thus, the log-likelihood is strictly concave in μ_k, and the stationary point is its unique maximizer for every fixed $K \succ 0$.
In particular, the maximum likelihood estimators of the mean vectors are independent of the feature dimension n.

Estimation of the Covariance Matrix.

Having obtained the maximum likelihood estimates of the mean vectors, define

$$r_i = x^{(i)} - \hat{\mu}_{y^{(i)}}$$

and the scatter matrix

$$\boxed{S = \sum_{i=1}^{m} r_i r_i^T.}$$

The terms depending on K are

$$\ell(K) = \frac{m}{2} \log\det K - \frac{1}{2} \sum_{i=1}^{m} r_i^T K r_i + C.$$

Since each quadratic form is a scalar, we have

$$\begin{aligned} r_i^T K r_i &= \operatorname{tr}(r_i^T K r_i) \\ &= \operatorname{tr}(K r_i r_i^T), \end{aligned}$$

where the second equality follows from the cyclic property of the trace.
Therefore,

$$\begin{aligned} \sum_{i=1}^{m} r_i^T K r_i &= \sum_{i=1}^{m} \operatorname{tr}(K r_i r_i^T) \\ &= \operatorname{tr}\left(K \sum_{i=1}^{m} r_i r_i^T \right) \\ &= \operatorname{tr}(KS). \end{aligned}$$

Hence,

$$\boxed{\ell(K) = \frac{m}{2} \log\det K - \frac{1}{2} \operatorname{tr}(KS) + C.}$$

We now differentiate with respect to the precision matrix K.
Recall the established matrix calculus identities

$$\nabla_A \operatorname{tr}(AB) = B^T$$

and

$$\nabla_A \det A = \det(A)(A^{-1})^T.$$

By the chain rule,

$$\begin{aligned} \nabla_A \log\det A &= \frac{1}{\det A} \nabla_A \det A \\ &= \frac{1}{\det A} \det(A)(A^{-1})^T \\ &= (A^{-1})^T. \end{aligned}$$

Thus,

$$\boxed{\nabla_A \log\det A = (A^{-1})^T.}$$

Applying these identities to the log-likelihood,

$$\begin{aligned} \nabla_K \ell &= \frac{m}{2} \nabla_K \log\det K - \frac{1}{2} \nabla_K \operatorname{tr}(KS) \\ &= \frac{m}{2} (K^{-1})^T - \frac{1}{2} S^T. \end{aligned}$$

Since K and S are symmetric,

$$(K^{-1})^T = K^{-1}, \qquad S^T = S.$$

Therefore,

$$\boxed{\nabla_K \ell = \frac{1}{2} \left(m K^{-1} - S \right).}$$

Applying the first-order optimality condition,

$$\nabla_K \ell = 0,$$

we obtain

$$mK^{-1} - S = 0,$$
$$mK^{-1} = S,$$
$$K^{-1} = \frac{1}{m} S.$$

Recalling that $K = \Sigma^{-1}$, we conclude that

$$\boxed{\hat{\Sigma} = \frac{1}{m} S.}$$

Substituting the definition of S,

$$\boxed{\hat{\Sigma} = \frac{1}{m} \sum_{i=1}^{m} \left(x^{(i)} - \hat{\mu}_{y^{(i)}} \right) \left(x^{(i)} - \hat{\mu}_{y^{(i)}} \right)^T.}$$

Verification of Global Maximality.
It remains to verify that the stationary point is a global maximizer.
Recall that

$$\ell(K) = \frac{m}{2} \log\det K - \frac{1}{2} \operatorname{tr}(KS) + C.$$

The function

$$K \longmapsto \log\det K$$

is strictly concave on the set of symmetric positive definite matrices.
Furthermore,

$$K \longmapsto \operatorname{tr}(KS)$$

is linear in K.
Consequently, $\ell(K)$ is strictly concave on the convex set of symmetric positive definite matrices.
By assumption, $S \succ 0$. Therefore,

$$K^* = mS^{-1}$$

exists and is positive definite.
Moreover,

$$\nabla_K \ell(K^*) = 0.$$

Since ℓ is strictly concave, this stationary point is its unique global maximizer.
Hence,

$$\boxed{\hat{\Sigma} = (K^*)^{-1} = \frac{1}{m} S.}$$

Conclusion.
Combining the results of parts (i) and (ii), the maximum likelihood estimators of the GDA parameters are

$$\boxed{\begin{aligned}
\hat{\phi} &= \frac{1}{m} \sum_{i=1}^{m} \mathbf{1} y^{(i)} = 1, \\
\hat{\mu}_0 &= \frac{\sum_{i=1}^{m} \mathbf{1} y^{(i)} = 0 x^{(i)}}{\sum_{i=1}^{m} \mathbf{1} y^{(i)} = 0}, \\
\hat{\mu}_1 &= \frac{\sum_{i=1}^{m} \mathbf{1} y^{(i)} = 1 x^{(i)}}{\sum_{i=1}^{m} \mathbf{1} y^{(i)} = 1}, \\
\hat{\Sigma} &= \frac{1}{m} \sum_{i=1}^{m} (x^{(i)} - \hat{\mu}_{y^{(i)}})(x^{(i)} - \hat{\mu}_{y^{(i)}})^T.
\end{aligned}}$$

Thus, the maximum likelihood estimators derived in the one-dimensional case extend to arbitrary finite dimensions, provided that the empirical residual covariance matrix is positive definite.

□

(e) [3 points] Coding problem.
In src/p01e_gda.py, fill in the code to calculate ϕ, μ_0, μ_1, and Σ, use these parameters to derive θ, and use the resulting GDA model to make predictions on the validation set. (Hint: you might want to rewrite θ_0 in the appropriate form by the difference-of-squares identity for symmetric matrix A: $u^T Au - vA^T v = (u+v)^T A(u-v)$)

(f) [5 points]
For Dataset 1, create a plot of the training data with x_1 on the horizontal axis, and x_2 on the vertical axis. To visualize the two classes, use a different symbol for examples $x^{(i)}$ with $y^{(i)} = 0$ than for those with $y^{(i)} = 1$.
On the same figure, plot the decision boundary found by logistic regression in part (b).
Make an identical plot with the decision boundary found by GDA in part (e).

(g) [5 points]
Repeat the steps in part (f) for Dataset 2.
On which dataset does GDA seem to perform worse than logistic regression?
Why might this be the case?

Solution.
For dataset one, Linear regression has obviouly done a better job in determining the decision boundary, while the decision boundaries for both models in dataset two are roughly the same. Recall Mahalanobis distance, defined by $d_E(x, \mu) = \sqrt{(x-\mu)^T(x-\mu)}$: In multivariate distribution(in particular, when d = 2, it's an ellipse), the level curve of the probability density function is just a dim(d) ellipsoid. Notice that here, the geometry of data2 is roughly a slant ellipse, making it valid and appropriate to assume Gaussian distribution of $X|Y = y_i$.

(h) [3 extra credit points]
For the dataset where GDA performed worse in parts (f) and (g), can you find a transformation of the $x^{(i)}$'s such that GDA performs significantly better?
What is this transformation?

Solution.
Box-Cox transformation: define

$$\boxed{T_\lambda(x) = \begin{cases} \dfrac{x^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log x, & \lambda = 0. \end{cases}}$$

Note: we tempararily skip this part, and we'll come back later if we have more time

2. [30 points] Incomplete, Positive-Only Labels

In this problem we will consider training binary classifiers in situations where we do not have full access to the labels.

In particular, we consider a scenario, which is not too infrequent in real life, where we have labels only for a subset of the positive examples. All the negative examples and the rest of the positive examples are unlabeled.

That is, we assume a dataset

$$\left\{(x^{(i)}, t^{(i)}, y^{(i)})\right\}_{i=1}^{m},$$

where

$$t^{(i)} \in \{0, 1\}$$

is the "true" label, and where

$$y^{(i)} = \begin{cases} 1, & x^{(i)} \text{ is labeled,} \\ 0, & \text{otherwise.} \end{cases}$$

All labeled examples are positive, which is to say

$$p(t^{(i)} = 1 \mid y^{(i)} = 1) = 1,$$

but unlabeled examples may be positive or negative.

Our goal in the problem is to construct a binary classifier h of the true label t, with only access to the partial labels y.

In other words, we want to construct h such that

$$h(x^{(i)}) \approx p(t^{(i)} = 1 \mid x^{(i)})$$

as closely as possible, using only x and y.

Real world example: Suppose we maintain a database of proteins which are involved in transmitting signals across membranes. Every example added to the database is involved in a signaling process, but there are many proteins involved in cross-membrane signaling which are missing from the database.

It would be useful to train a classifier to identify proteins that should be added to the database.

In our notation, each example $x^{(i)}$ corresponds to a protein, $y^{(i)} = 1$ if the protein is in the database and 0 otherwise, and $t^{(i)} = 1$ if the protein is involved in a cross-membrane signaling process and thus should be added to the database, and 0 otherwise.

(a) [5 points]

Suppose that each $y^{(i)}$ and $x^{(i)}$ are conditionally independent given $t^{(i)}$:

$$p(y^{(i)} = 1 \mid t^{(i)} = 1, x^{(i)}) = p(y^{(i)} = 1 \mid t^{(i)} = 1).$$

Note this is equivalent to saying that labeled examples were selected uniformly at random from the set of positive examples.

Prove that the probability of an example being labeled differs by a constant factor from the probability of an example being positive.

That is, show that

$$p(t^{(i)} = 1 \mid x^{(i)}) = \frac{p(y^{(i)} = 1 \mid x^{(i)})}{\alpha}$$

for some

$$\alpha \in \mathbb{R}.$$

Solution.

Let's first briefly review the concept of Conditional Chain Rule.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $A, B, C \in \mathcal{F}$ be events such that $\mathbb{P}(B \cap C) > 0$. Then

$$\boxed{\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid B \cap C)\,\mathbb{P}(B \mid C).}$$

The notation $\mathbb{P}(A \mid B, C)$ is shorthand for $\mathbb{P}(A \mid B \cap C)$; the comma denotes simultaneous conditioning on both events.

Theorem (Conditional Law of Total Probability). Let $\{B_i\}_{i=1}^n$ be a finite measurable partition of Ω(total space), and let $C \in \mathcal{F}$ satisfy $\mathbb{P}(C) > 0$. Then, for any event $A \in \mathcal{F}$,

$$\boxed{\mathbb{P}(A \mid C) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i \cap C)\, \mathbb{P}(B_i \mid C).}$$

Terms with $\mathbb{P}(B_i \cap C) = 0$ are understood to contribute zero to the sum.

Proof. Since $\{B_i\}_{i=1}^n$ partitions Ω, the events $\{A \cap B_i \cap C\}_{i=1}^n$ are pairwise disjoint, and

$$A \cap C = \bigcup_{i=1}^{n} (A \cap B_i \cap C).$$

Consequently,

$$\begin{aligned}
\mathbb{P}(A \mid C) &= \frac{\mathbb{P}(A \cap C)}{\mathbb{P}(C)} \\
&= \sum_{i=1}^{n} \frac{\mathbb{P}(A \cap B_i \cap C)}{\mathbb{P}(C)} \\
&= \sum_{i=1}^{n} \underbrace{\frac{\mathbb{P}(A \cap B_i \cap C)}{\mathbb{P}(B_i \cap C)}}_{\mathbb{P}(A \mid B_i \cap C)} \underbrace{\frac{\mathbb{P}(B_i \cap C)}{\mathbb{P}(C)}}_{\mathbb{P}(B_i \mid C)} \\
&= \sum_{i=1}^{n} \mathbb{P}(A \mid B_i \cap C)\, \mathbb{P}(B_i \mid C).
\end{aligned}$$

□

Now back to the original problem. We are motivated to inspect the relationship between $p(t^{(i)}|x^{(i)})$ and $p(y^{(i)} = 1|x^{(i)})$.
We first formalize these variables. Let X be a feature vector, $T \in \{0, 1\}$ a latent binary label, and $Y \in \{0, 1\}$ an observed label. Conditioning on $X = x$ and partitioning according to the possible values of T, we obtain

$$\boxed{\begin{aligned}\mathbb{P}(Y = 1 \mid X = x) = \mathbb{P}(Y = 1 \mid T = 1, X = x)\mathbb{P}(T = 1 \mid X = x) \\ + \mathbb{P}(Y = 1 \mid T = 0, X = x)\mathbb{P}(T = 0 \mid X = x).\end{aligned}}$$

For continuous X, this identity is interpreted using regular conditional probabilities rather than conditioning directly on the zero-probability event $\{X = x\}$. It holds for almost every x with respect to the distribution of X.
We know that negative examples are never labeled:

$$\mathbb{P}(Y = 1 \mid T = 0, X = x) = 0.$$

Furthermore, we know that labeling probability among true positive examples is independent of X:

$$\mathbb{P}(Y = 1 \mid T = 1, X = x) = \mathbb{P}(Y = 1 \mid T = 1) =: \alpha.$$

Then the conditional law of total probability gives

$$\mathbb{P}(Y = 1 \mid X = x) = \alpha\, \mathbb{P}(T = 1 \mid X = x).$$

Therefore, provided $\alpha > 0$,

$$\boxed{\mathbb{P}(T = 1 \mid X = x) = \frac{\mathbb{P}(Y = 1 \mid X = x)}{\alpha}.}$$

(b) [5 points]
Suppose we want to estimate α using a trained classifier h and a held-out validation set V.
Let V_+ be the set of labeled (and hence positive) examples in V, given by

$$V_+ = \left\{ x^{(i)} \in V \mid y^{(i)} = 1 \right\}.$$

Assuming that

$$h(x^{(i)}) \approx p(y^{(i)} = 1 \mid x^{(i)})$$

for all examples $x^{(i)}$, show that

$$h(x^{(i)}) \approx \alpha$$

for all

$$x^{(i)} \in V_+.$$

You may assume that

$$p(t^{(i)} = 1 \mid x^{(i)}) \approx 1$$

when

$$x^{(i)} \in V_+.$$

Solution.

From part (a), we have established that

$$P(Y = 1 \mid X = x) = \alpha P(T = 1 \mid X = x),$$

where

$$\alpha = P(Y = 1 \mid T = 1).$$

By assumption, the classifier h satisfies

$$h(x) \approx P(Y = 1 \mid X = x).$$

Furthermore, for every labeled positive example $x \in V_+$, we are given that

$$P(T = 1 \mid X = x) \approx 1.$$

Consequently,

$$\begin{aligned} h(x) &\approx P(Y = 1 \mid X = x) \\ &= \alpha P(T = 1 \mid X = x) \\ &\approx \alpha. \end{aligned}$$

Therefore,

$$\boxed{h(x^{(i)}) \approx \alpha, \qquad \forall x^{(i)} \in V_+.}$$

□

(c) [5 points] Coding problem.
The following three problems will deal with a dataset which we have provided in the following files:

data/ds3_{train,valid,test}.csv.

Each file contains the following columns: x_1, x_2, y, and t. As in Problem 1, there is one example per row.
First we will consider the ideal case, where we have access to the true t-labels for training.
In src/p02cde_posonly.py, write a logistic regression classifier that uses x_1 and x_2 as input features, and train it using the t-labels.
You can ignore the y-labels for this part.
Output the trained model's predictions on the test set to the file specified in the code.

(d) [5 points] Coding problem.
We now consider the case where the t-labels are unavailable, so you only have access to the y-labels at training time.
Add to your code in p02cde_posonly.py to re-train the classifier (still using x_1 and x_2 as input features), but using the y-labels only.

(e) [10 points] Coding problem.
Using the validation set, estimate the constant α by averaging your classifier's predictions over all labeled examples in the validation set:

$$\alpha \approx \frac{1}{|V_+|} \sum_{x^{(i)} \in V_+} h(x^{(i)}).$$

Add code in src/p02cde_posonly.py to rescale your classifier's predictions from part (d) using the estimated value for α.
Finally, using a threshold of

$$p(t^{(i)} = 1 \mid x^{(i)}) = 0.5,$$

make three separate plots with the decision boundaries from parts (c)–(e) plotted on top of the test set.
Plot x_1 on the horizontal axis and x_2 on the vertical axis, and use two different symbols for the positive ($t^{(i)} = 1$) and negative ($t^{(i)} = 0$) examples.
In each plot, indicate the separating hyperplane with a red line.

Remark: We saw that the true probability $p(t \mid x)$ was only a constant factor away from $p(y \mid x)$.
This means, if our task is to only rank examples (i.e. sort them) in a particular order, then in fact we do not even need to estimate α.
The rank based on $p(y \mid x)$ will agree with the rank based on $p(t \mid x)$.

3. [25 points] Poisson Regression

(a) [5 points]
Consider the Poisson distribution parameterized by λ:

$$p(y;\lambda) = \frac{e^{-\lambda}\lambda^y}{y!}.$$

Show that the Poisson distribution is in the exponential family, and clearly state the values for

$$b(y), \qquad \eta, \qquad T(y), \qquad a(\eta).$$

Solution.

Rewriting the probability mass function, we obtain

$$\begin{aligned} p(y;\lambda) &= \exp\left(\ln \frac{e^{-\lambda}\lambda^y}{y!}\right) \\ &= \exp\left(-\lambda + y\ln\lambda - \ln(y!)\right) \\ &= \frac{1}{y!}\exp\left(y\ln\lambda - \lambda\right). \end{aligned}$$

Define the natural parameter

$$\eta = \ln\lambda \quad \Longleftrightarrow \quad \lambda = e^{\eta}.$$

Substituting this into the expression above gives

$$p(y;\eta) = \frac{1}{y!}\exp\left(\eta y - e^{\eta}\right).$$

Comparing with the exponential family representation, we identify

$$\boxed{\begin{aligned} b(y) &= \frac{1}{y!}, \\ \eta &= \ln\lambda, \\ T(y) &= y, \\ a(\eta) &= e^{\eta}. \end{aligned}}$$

Therefore, the Poisson distribution belongs to the exponential family. □

(b) [3 points]
Consider performing regression using a GLM model with a Poisson response variable.
What is the canonical response function for the family?
You may use the fact that a Poisson random variable with parameter λ has mean λ.

Solution.

From part (a), the Poisson distribution belongs to the exponential family with natural parameter $\eta = \log\lambda$ and log-partition function

$$a(\eta) = e^{\eta}.$$

Recall that the canonical response function of a GLM is given by

$$h_\theta(x) = \mathbb{E}[Y \mid X = x;\theta].$$

Since $T(y) = y$, by using the formula proven in Q4

$$\mathbb{E}[Y \mid X = x;\theta] = \frac{\partial a(\eta)}{\partial \eta} = e^{\eta}.$$

Under the canonical link assumption $\eta = \theta^T x$, it follows that

$$\boxed{h_\theta(x) = e^{\theta^T x}.}$$

Therefore, the canonical response function for Poisson regression is the exponential function. □

(c) [7 points]
For a training set

$$\left\{(x^{(i)}, y^{(i)});\ i = 1, \ldots, m\right\},$$

let the log-likelihood of an example be

$$\log p(y^{(i)} \mid x^{(i)}; \theta).$$

By taking the derivative of the log-likelihood with respect to θ_j, derive the stochastic gradient ascent update rule for learning using a GLM model with Poisson responses y and the canonical response function.

Solution.

For a single training example $(x^{(i)}, y^{(i)})$, the Poisson GLM assumes that

$$Y \mid X = x^{(i)}; \theta \sim \text{Poisson}(\lambda_i), \qquad \lambda_i = e^{\theta^T x^{(i)}}.$$

Thus, the conditional probability mass function is

$$p(y^{(i)} \mid x^{(i)}; \theta) = \frac{1}{y^{(i)}!} \exp\left(y^{(i)} \theta^T x^{(i)} - e^{\theta^T x^{(i)}}\right).$$

Define the single-example likelihood function by

$$L_i(\theta) = p(y^{(i)} \mid x^{(i)}; \theta).$$

Taking the logarithm, we obtain the log-likelihood

$$\begin{aligned} \ell_i(\theta) &= \log L_i(\theta) \\ &= -\log(y^{(i)}!) + y^{(i)} \theta^T x^{(i)} - e^{\theta^T x^{(i)}}. \end{aligned}$$

Taking the gradient of the log-likelihood with respect to θ, we obtain

$$\begin{aligned} \nabla_\theta \ell_i(\theta) &= \nabla_\theta \left(y^{(i)} \theta^T x^{(i)} - e^{\theta^T x^{(i)}}\right) \\ &= y^{(i)} x^{(i)} - e^{\theta^T x^{(i)}} x^{(i)} \\ &= \boxed{\left(y^{(i)} - e^{\theta^T x^{(i)}}\right) x^{(i)}}. \end{aligned}$$

Since we seek to maximize the log-likelihood, we apply stochastic gradient ascent with learning rate $\alpha > 0$:

$$\theta^{(t+1)} = \theta^{(t)} + \alpha \nabla_\theta \ell_i(\theta^{(t)}).$$

Substituting the gradient yields

$$\boxed{\theta^{(t+1)} = \theta^{(t)} + \alpha \left(y^{(i)} - e^{(\theta^{(t)})^T x^{(i)}}\right) x^{(i)}.}$$

Finally, recalling that the canonical response function for Poisson regression is

$$h_\theta(x) = e^{\theta^T x},$$

we can express the stochastic gradient ascent update as

$$\boxed{\theta^{(t+1)} = \theta^{(t)} + \alpha \left(y^{(i)} - h_{\theta^{(t)}}(x^{(i)})\right) x^{(i)}.}$$

□

(d) [7 points] Coding problem.
Consider a website that wants to predict its daily traffic.
The website owners have collected a dataset of past traffic to their website, along with some features which they think are useful in predicting the number of visitors per day.
The dataset is split into train/valid/test sets and follows the same format as Datasets 1–3:

data/ds4_{train,valid}.csv.

We will apply Poisson regression to model the number of visitors per day.
Note that applying Poisson regression in particular assumes that the data follows a Poisson distribution whose natural parameter is a linear combination of the input features:

$$\eta = \theta^T x.$$

In src/p03d_poisson.py, implement Poisson regression for this dataset and use gradient ascent to maximize the log-likelihood of θ.

4. [15 points] Convexity of Generalized Linear Models

In this question we will explore and show some nice properties of Generalized Linear Models, specifically those related to their use of Exponential Family distributions to model the output.

Most commonly, GLMs are trained by using the negative log-likelihood (NLL) as the loss function. This is mathematically equivalent to Maximum Likelihood Estimation; maximizing the log-likelihood is equivalent to minimizing the negative log-likelihood.

In this problem, our goal is to show that the NLL loss of a GLM is a convex function with respect to the model parameters.

As a reminder, this is convenient because a convex function is one for which any local minimum is also a global minimum.

To recap, an exponential family distribution is one whose probability density can be represented as

$$p(y;\eta) = b(y)\exp\left(\eta^T T(y) - a(\eta)\right),$$

where η is the natural parameter of the distribution.

Moreover, in a Generalized Linear Model, η is modeled as

$$\theta^T x,$$

where

$$x \in \mathbb{R}^n$$

are the input features of the example, and

$$\theta \in \mathbb{R}^n$$

are learnable parameters.

In order to show that the NLL loss is convex for GLMs, we break down the process into sub-parts, and approach them one at a time.

Our approach is to show that the second derivative (i.e. Hessian) of the loss with respect to the model parameters is Positive Semi-Definite (PSD) at all values of the model parameters.

We will also show some nice properties of Exponential Family distributions as intermediate steps.

For the sake of convenience we restrict ourselves to the case where η is a scalar.

Assume

$$p(Y \mid X;\theta) \sim \text{ExponentialFamily}(\eta),$$

where

$$\eta \in \mathbb{R}$$

is a scalar, and

$$T(y) = y.$$

This makes the exponential family representation take the form

$$p(y;\eta) = b(y)\exp(\eta y - a(\eta)).$$

(a) [5 points]

Derive an expression for the mean of the distribution.

Show that

$$\mathbb{E}[Y \mid X;\theta]$$

can be represented as the gradient of the log-partition function a with respect to the natural parameter η.

Hint: Start with observing that

$$\frac{\partial}{\partial\eta}\int p(y;\eta)\,dy = \int \frac{\partial}{\partial\eta} p(y;\eta)\,dy.$$

Solution.

By the definition of conditional expectation,

$$\mathbb{E}[Y \mid X = x;\theta] = \int_{\mathcal{Y}} y\, p(y;\eta)\,dy.$$

First, differentiating $p(y;\eta)$ with respect to η, we obtain

$$\begin{aligned}\frac{\partial}{\partial\eta}p(y;\eta) &= b(y)\exp\left(\eta y - a(\eta)\right)\left(y - \frac{\partial a}{\partial\eta}\right)\\ &= p(y;\eta)\left(y - \frac{\partial a}{\partial\eta}\right)\\ &= y\,p(y;\eta) - \frac{\partial a}{\partial\eta}p(y;\eta).\end{aligned}$$

Rearranging,

$$\boxed{y\,p(y;\eta) = \frac{\partial}{\partial\eta}p(y;\eta) + \frac{\partial a}{\partial\eta}p(y;\eta).}$$

Substituting into the expression for the conditional expectation,

$$\begin{aligned}\mathbb{E}[Y \mid X = x;\theta] &= \int_{\mathcal{Y}} y\,p(y;\eta)\,dy\\ &= \int_{\mathcal{Y}} \frac{\partial}{\partial\eta}p(y;\eta)\,dy + \frac{\partial a}{\partial\eta}\int_{\mathcal{Y}} p(y;\eta)\,dy\\ &= \frac{\partial}{\partial\eta}\int_{\mathcal{Y}} p(y;\eta)\,dy + \frac{\partial a}{\partial\eta}\int_{\mathcal{Y}} p(y;\eta)\,dy\\ &= \frac{\partial}{\partial\eta}(1) + \frac{\partial a}{\partial\eta}\cdot 1\\ &= \frac{\partial a}{\partial\eta}.\end{aligned}$$

Here, differentiation under the integral sign is justified by the regularity conditions of the exponential family, and we have used the normalization property

$$\int_{\mathcal{Y}} p(y;\eta)\,dy = 1.$$

Therefore,

$$\boxed{\mathbb{E}[Y \mid X = x;\theta] = a'(\eta) = a'(\theta^T x).}$$

Hence, the conditional mean is given by the first derivative of the log-partition function with respect to the natural parameter. □

(b) [5 points]
Next, derive an expression for the variance of the distribution.
In particular, show that

$$\mathrm{Var}(Y \mid X;\theta)$$

can be expressed as the derivative of the mean with respect to η, i.e. the second derivative of the log-partition function $a(\eta)$ with respect to the natural parameter η.

Solution.

Recall that the exponential family distribution with $T(y) = y$ is given by

$$p(y;\eta) = b(y)\exp\left(\eta y - a(\eta)\right), \qquad \eta = \theta^T x.$$

By the definition of conditional variance,

$$\begin{aligned}\mathrm{Var}(Y \mid X = x;\theta) &= \mathbb{E}\left[(Y - \mathbb{E}[Y \mid X = x;\theta])^2 \mid X = x;\theta\right]\\ &= \mathbb{E}[Y^2 \mid X = x;\theta] - (\mathbb{E}[Y \mid X = x;\theta])^2.\end{aligned}$$

From part (a), we have established that

$$\mathbb{E}[Y \mid X = x;\theta] = a'(\eta).$$

Thus, it suffices to show that

$$a''(\eta) = \mathbb{E}[Y^2 \mid X = x; \theta] - (a'(\eta))^2 .$$

Differentiating the result from part (a) with respect to the natural parameter η, we obtain

$$\frac{\partial}{\partial \eta}\mathbb{E}[Y \mid X = x; \theta] = a''(\eta).$$

On the other hand, by the definition of conditional expectation,

$$\mathbb{E}[Y \mid X = x; \theta] = \int_{\mathcal{Y}} y\, p(y; \eta)\, dy.$$

Assuming the regularity conditions that justify differentiation under the integral sign,

$$\begin{aligned}\frac{\partial}{\partial \eta}\mathbb{E}[Y \mid X = x; \theta] &= \frac{\partial}{\partial \eta}\int_{\mathcal{Y}} y\, p(y; \eta)\, dy \\ &= \int_{\mathcal{Y}} \frac{\partial}{\partial \eta}\left(y\, p(y; \eta)\right)\, dy.\end{aligned}$$

Using the exponential family representation,

$$\begin{aligned}\frac{\partial}{\partial \eta}\left(y\, p(y; \eta)\right) &= \frac{\partial}{\partial \eta}\left(y\, b(y) e^{\eta y - a(\eta)}\right) \\ &= y\, b(y) e^{\eta y - a(\eta)}\left(y - a'(\eta)\right) \\ &= y^2 p(y; \eta) - a'(\eta) y\, p(y; \eta).\end{aligned}$$

Substituting this expression into the integral yields

$$\begin{aligned}a''(\eta) &= \int_{\mathcal{Y}} \left(y^2 p(y; \eta) - a'(\eta) y\, p(y; \eta)\right)\, dy \\ &= \int_{\mathcal{Y}} y^2 p(y; \eta)\, dy - a'(\eta)\int_{\mathcal{Y}} y\, p(y; \eta)\, dy \\ &= \mathbb{E}[Y^2 \mid X = x; \theta] - a'(\eta)\mathbb{E}[Y \mid X = x; \theta] \\ &= \mathbb{E}[Y^2 \mid X = x; \theta] - \left(\mathbb{E}[Y \mid X = x; \theta]\right)^2 \\ &= \mathrm{Var}(Y \mid X = x; \theta).\end{aligned}$$

Therefore,

$$\boxed{\mathrm{Var}(Y \mid X = x; \theta) = \frac{\partial}{\partial \eta}\mathbb{E}[Y \mid X = x; \theta] = \frac{\partial^2 a(\eta)}{\partial \eta^2} = a''(\theta^T x).}$$

Hence, the conditional variance is equal to the second derivative of the log-partition function with respect to the natural parameter. □

(c) [5 points]
Finally, write out the loss function

$$\ell(\theta),$$

the NLL of the distribution, as a function of θ.
Then calculate the Hessian of the loss with respect to θ, and show that it is always PSD.
This concludes the proof that NLL loss of GLM is convex.
Hint: Use the chain rule of calculus along with the results of the previous parts to simplify your derivations.

Solution.

For a single training example (x, y), the likelihood is

$$L(\theta) = p(y \mid x; \theta),$$

and the negative log-likelihood is defined as

$$\ell(\theta) = -\log L(\theta).$$

Recall that the exponential family distribution, with $T(y) = y$ and natural parameter $\eta = \theta^T x$, is given by

$$p(y \mid x; \theta) = b(y) \exp\left(y\theta^T x - a(\theta^T x)\right).$$

Substituting this expression into the NLL, we obtain

$$\begin{aligned}\ell(\theta) &= -\log\left[b(y) \exp\left(y\theta^T x - a(\theta^T x)\right)\right] \\ &= -\log b(y) - y\theta^T x + a(\theta^T x).\end{aligned}$$

Taking the gradient with respect to θ and applying the chain rule,

$$\begin{aligned}\nabla_\theta \ell(\theta) &= -y\nabla_\theta(\theta^T x) + \nabla_\theta a(\theta^T x) \\ &= -yx + a'(\theta^T x)x \\ &= \left(a'(\theta^T x) - y\right)x.\end{aligned}$$

Differentiating once more, we obtain the Hessian:

$$\boxed{\nabla_\theta^2 \ell(\theta) = a''(\theta^T x)xx^T.}$$

We now prove that this Hessian is positive semidefinite.
By definition, a symmetric matrix H is positive semidefinite if

$$v^T H v \geq 0, \qquad \forall v \in \mathbb{R}^n.$$

Let $v \in \mathbb{R}^n$ be arbitrary. Then

$$\begin{aligned}v^T \nabla_\theta^2 \ell(\theta) v &= v^T \left(a''(\theta^T x)xx^T\right) v \\ &= a''(\theta^T x) v^T x x^T v \\ &= a''(\theta^T x)(x^T v)^T (x^T v) \\ &= a''(\theta^T x)(x^T v)^2.\end{aligned}$$

From part (b), we have established that

$$a''(\eta) = \mathrm{Var}(Y \mid X = x; \theta) \geq 0.$$

Moreover, since $x^T v \in \mathbb{R}$,

$$(x^T v)^2 \geq 0.$$

Consequently,

$$\boxed{v^T \nabla_\theta^2 \ell(\theta) v = a''(\theta^T x)(x^T v)^2 \geq 0.}$$

Since v was arbitrary, it follows that

$$\boxed{\nabla_\theta^2 \ell(\theta) \succeq 0.}$$

Hence, the single-example negative log-likelihood $\ell(\theta)$ is convex on its natural convex parameter domain.
For a dataset of m conditionally independent training examples, the total negative log-likelihood is

$$J(\theta) = \sum_{i=1}^m \ell_i(\theta).$$

Its Hessian is

$$\nabla_\theta^2 J(\theta) = \sum_{i=1}^m a''(\theta^T x^{(i)}) x^{(i)} (x^{(i)})^T.$$

Since each summand is positive semidefinite, their sum is also positive semidefinite:

$$\boxed{\nabla_\theta^2 J(\theta) \succeq 0.}$$

Therefore, the negative log-likelihood loss of a canonical GLM is convex on its convex parameter domain. □

Remark: The main takeaways from this problem are:

- Any GLM model is convex in its model parameters.
- The exponential family of probability distributions are mathematically nice. Whereas calculating mean and variance of distributions in general involves integrals, we can calculate them using derivatives for exponential family distributions.

5. [25 points] Locally weighted linear regression

(a) [10 points]

Consider a linear regression problem in which we want to "weight" different training examples differently. Specifically, suppose we want to minimize

$$J(\theta) = \frac{1}{2}\sum_{i=1}^{m} w^{(i)} \left(\theta^T x^{(i)} - y^{(i)}\right)^2.$$

In class, we worked out what happens for the case where all the weights (the $w^{(i)}$'s) are the same. In this problem, we will generalize some of those ideas to the weighted setting.

i. [2 points]

Show that $J(\theta)$ can also be written

$$J(\theta) = (X\theta - y)^T W (X\theta - y)$$

for an appropriate matrix W, and where X and y are as defined in class. Clearly specify the value of each element of the matrix W.

Solution.

Suppose we have m training examples, each with n features. Define the design matrix $X \in \mathbb{R}^{m\times n}$, the label vector $y \in \mathbb{R}^m$, and the parameter vector $\theta \in \mathbb{R}^n$ by

$$X = \begin{pmatrix} (x^{(1)})^T \\ (x^{(2)})^T \\ \vdots \\ (x^{(m)})^T \end{pmatrix}, \qquad y = \begin{pmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{pmatrix}.$$

Observe that

$$X\theta - y = \begin{pmatrix} \theta^T x^{(1)} - y^{(1)} \\ \theta^T x^{(2)} - y^{(2)} \\ \vdots \\ \theta^T x^{(m)} - y^{(m)} \end{pmatrix}.$$

Now define the diagonal weight matrix

$$\boxed{W = \frac{1}{2}\begin{pmatrix} w^{(1)} & 0 & \cdots & 0 \\ 0 & w^{(2)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w^{(m)} \end{pmatrix} \in \mathbb{R}^{m\times m}.}$$

Equivalently, its entries are

$$W_{ij} = \begin{cases} \frac{1}{2} w^{(i)}, & i = j, \\ 0, & i \neq j. \end{cases}$$

Expanding the quadratic form, we obtain

$$\begin{aligned}(X\theta - y)^T W (X\theta - y) &= \frac{1}{2}\sum_{i=1}^{m} w^{(i)} \left(\theta^T x^{(i)} - y^{(i)}\right)^2 \\ &= J(\theta).\end{aligned}$$

Therefore,

$$\boxed{J(\theta) = (X\theta - y)^T W (X\theta - y).}$$

ii. [4 points]

If all the $w^{(i)}$'s equal 1, then we saw in class that the normal equation is

$$X^T X\theta = X^T y,$$

and that the value of θ that minimizes $J(\theta)$ is given by

$$(X^T X)^{-1} X^T y.$$

By finding the derivative

$$\nabla_\theta J(\theta)$$

and setting that to zero, generalize the normal equation to this weighted setting, and give the new value of θ that minimizes $J(\theta)$ in closed form as a function of X, W, and y.

Solution.

From Part (i), the weighted least-squares objective is

$$J(\theta) = (X\theta - y)^T W (X\theta - y),$$

where W is a symmetric diagonal matrix.
Expanding the quadratic form gives

$$\begin{aligned} J(\theta) &= (\theta^T X^T - y^T) W (X\theta - y) \\ &= \theta^T X^T W X \theta - \theta^T X^T W y - y^T W X \theta + y^T W y. \end{aligned}$$

Since $W = W^T$, the matrix $X^T W X$ is symmetric. Recall the matrix differentiation identities

$$\nabla_\theta(\theta^T A \theta) = (A + A^T)\theta, \qquad \nabla_\theta(\theta^T b) = b.$$

Taking the gradient with respect to θ, we obtain

$$\begin{aligned} \nabla_\theta J(\theta) &= 2X^T W X \theta - X^T W y - X^T W^T y \\ &= 2X^T W X \theta - 2X^T W y \\ &= 2X^T W (X\theta - y). \end{aligned}$$

Setting the gradient equal to zero yields

$$2X^T W (X\theta - y) = 0.$$

Consequently,

$$\boxed{X^T W X \theta = X^T W y.}$$

This is the weighted normal equation.
Assuming $X^T W X$ is invertible, multiplying both sides by $(X^T W X)^{-1}$ gives

$$\boxed{\theta^* = (X^T W X)^{-1} X^T W y.}$$

Furthermore, since W is positive semidefinite,

$$\nabla^2_\theta J(\theta) = 2X^T W X \succeq 0.$$

Thus, J is convex, and the stationary point θ^* is a global minimizer.

iii. [4 points]
Suppose we have a dataset

$$\left\{(x^{(i)}, y^{(i)});\ i = 1, \ldots, m\right\}$$

of m independent examples, but we model the $y^{(i)}$'s as drawn from conditional distributions with different levels of variance $(\sigma^{(i)})^2$.
Specifically, assume the model

$$p(y^{(i)} \mid x^{(i)}; \theta) = \frac{1}{\sqrt{2\pi}\sigma^{(i)}} \exp\left(-\frac{\left(y^{(i)} - \theta^T x^{(i)}\right)^2}{2(\sigma^{(i)})^2}\right).$$

That is, each $y^{(i)}$ is drawn from a Gaussian distribution with mean

$$\theta^T x^{(i)}$$

and variance

$$(\sigma^{(i)})^2,$$

where the $\sigma^{(i)}$'s are fixed, known constants.
Show that finding the maximum likelihood estimate of θ reduces to solving a weighted linear regression problem.
State clearly what the $w^{(i)}$'s are in terms of the $\sigma^{(i)}$'s.

(b) [10 points] Coding problem.
We will now consider the following dataset, where $x^{(i)}$ is one-dimensional:

data/ds5_{train,valid,test}.csv.

In src/p05b_lwr.py, implement locally weighted linear regression using the normal equations you derived in part (a), and using

$$w^{(i)} = \exp\left(-\frac{\|x^{(i)} - x\|_2^2}{2\tau^2}\right).$$

Train your model on the train split using

$$\tau = 0.5,$$

then run your model on the valid split and report the mean squared error (MSE).
Finally, plot your model's predictions on the validation set. Plot the training set with blue x markers and the validation set with red o markers.
Does the model seem to be under- or overfitting?

(c) [5 points] Coding problem.
We will now tune the hyperparameter τ.
In src/p05c_tau.py, find the MSE value of your model on the validation set for each of the values of τ specified in the code.
For each τ, plot your model's predictions on the validation set in the format described in part (b).
Report the value of τ which achieves the lowest MSE on the valid split, and finally report the MSE on the test split using this τ-value.